

The Hong Kong Polytechnic University

Subject Description Form

Subject Code	AP620
Subject Title	Atomistic View of Matter: Modeling & Simulation
Credit Value	3
Level	6
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	This course will teach the physics that governs materials at the atomic scale and relate these processes to the macroscopic world.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a) design, perform and analyze computer experiments using electronic and atomistic simulation techniques appropriate for the problem at hand; b) extract materials properties from the simulations; c) recognize the approximations and estimate the level of accuracy to be expected from each modeling technique, and d) critically read the current scientific literature on computational modeling and simulation of materials.
Subject Synopsis/ Indicative Syllabus	The quantum mechanics of bonding: Atoms, molecules, molecular orbital theory, linear combination of atomic orbitals, variational principle Electronic structure calculations: Hartree, Hartree-Fock, post-Hartree-Fock, density functional theory and beyond, Koopmans theorem, modeling crystals Molecular dynamics simulations: Numerical integrators, neighbor list, spatial decomposition, interatomic potentials, parameterization, mean square displacement, auto-correlation, nudged elastic band Classical and statistical mechanics: thermodynamical ensembles, Boltzmann equation, equipartition energy, fluctuations, normal modes analysis, phonons, harmonic oscillator, specific heat, thermostats, barostats Advanced techniques: coarse grain simulations, atomistic electrochemistry, multi-scale modeling
Teaching/Learning Methodology	The course will make extensive use of cloud computing to enhance the learning using nanoHUB.org as well as its features for collaboration and publishing. Students will perform <i>ab initio</i> calculations using density functional theory and other advanced techniques as well as molecular dynamics simulations. Lecture: The concepts will be presented in class including some illustrative

	examples and live simulations. Homework assignments will be provided periodically for the students to apply the concepts developed in class and to solve problems of current research.					
	Computer laboratory: During laboratory sessions, students will perform atomistic simulations individually or in groups to gain a deeper understanding of topics related to the lectures. These sessions aim at developing the students’ intuition and critical thinking upon discussion and exchange of ideas in order to engage in the proposed activities.					
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)			
			a	b	c	d
	(1) Homework	20	✓	✓	✓	
	(2) Computer laboratory	20	✓	✓	✓	✓
	(2) Final project	60	✓	✓	✓	✓
	Total	100 %				
	Continuous assessment includes participation in the computer laboratory sessions and homework assignments in order to check students’ progress throughout the semester. Homework assignments will be graded as either complete or incomplete; in order to receive a passing grade students should not have more than one incomplete homework assignment. Computer labs will be graded.					
The final project consists of a take-home exam and will be assigned during the second half of the semester. Students will utilize the knowledge gained in the class to design and perform atomistic simulations in order to tackle a problem of current research.						
Student Study Effort Expected	Class contact:					
	• Lecture			27 Hrs.		
	Other student study effort:					
	• Self-study			81 Hrs.		
	• Computer Laboratory			12 Hrs.		
	Total student study effort			120 Hrs.		
Reading List and References	• “Modern Quantum Chemistry” - Attila Szabo & Neil S. Ostlund					
	• “Electronic Structure and the Properties of Solids” - Walter A. Harrison					
	• “Computational Physics” - J. M. Thijssen					

	• “From Atoms to Materials: Predictive Theory and Simulations” - Alejandro Strachan (2014), https://nanohub.org/courses/FATM
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